

Master Helm Hooks: Automate Tasks Before and After Deployments

Helm hooks let you run custom scripts like database migrations or alerts at specific stages of a deployment. Learning them prevents broken deployments and keeps your Kubernetes releases smooth and predictable.

What you'll learn:

- 1 Run Jobs Before App Starts (Pre-install Hook)
- 2 Send Alerts After Deploy (Post-install Hook)
- 3 Control Hook Sequence (Hook Weight)
- 4 Clean Up Pods Automatically (Delete Policy)

Aman Raj Singh

Cloud Operations Engineer @ iManage

linkedin.com/in/mrmanrajsingh

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Run Jobs Before App Starts (Pre-install Hook)

A pre-install hook creates a temporary pod that completes its job before any main application components are created. If this job fails, Helm stops the entire deployment, protecting your cluster from running an app with missing dependencies.

```
apiVersion: batch/v1
kind: Job
metadata:
  name: db-migrate
  annotations:
    "helm.sh/hook": pre-install
```

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Send Alerts After Deploy (Post-install Hook)

Post-install hooks run immediately after all application resources are loaded into Kubernetes. This is ideal for triggering external notifications, updating metric dashboards, or warming up caches once the system is ready.

```
apiVersion: batch/v1
kind: Job
metadata:
  name: send-slack-notification
  annotations:
    "helm.sh/hook": post-install
```

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Control Hook Sequence (Hook Weight)

When you have multiple hooks attached to the same stage, Helm runs them in ascending order based on their weight. Lower numbers run first, so a weight of -5 will execute before a weight of 0.

```
$metadata:
  annotations:
    "helm.sh/hook": pre-upgrade
    "helm.sh/hook-weight": "-5"
```

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Clean Up Pods Automatically (Delete Policy)

By default, pods created by Helm hooks stick around after finishing. Using a delete policy tells Helm to automatically remove hook resources when they succeed or get replaced.

```
metadata:
  annotations:
    "helm.sh/hook": pre-install
    "helm.sh/hook-delete-policy": hook-succeeded,before-hook-creation
```

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See Key Takeaways on next page >> linkedin.com/in/mrmanrajsingh

Key Takeaways

- + Always set hook-delete-policy to keep your namespace clean.
- + Use negative weights like '-5' to force critical pre-checks to run first.
- + Keep hook scripts idempotent so re-running them causes no harm.
- + Set an activeDeadlineSeconds on jobs so stuck hooks time out.
- + Test hook failures in a local cluster to verify safety behavior.

Watch Out For

- ! Forgetting delete policies, causing dozens of dead pods to accumulate.
- ! Leaving off timeouts, making Helm wait endlessly on frozen jobs.
- ! Putting main application logic into hooks instead of regular deployments.

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